Imbalance correction reverses CNN-transformer rankings in melanoma classification: a controlled benchmark with fairness and explanation analysis

Nursultan Zhanat
Supervisor: Prof. Sun Baoshan

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Abstract

Deep learning models for melanoma detection are trained on imbalanced, predominantly light-skinned data. We ask whether modern imbalance correction changes the ranking of CNN and transformer architectures, and whether the resulting models perform fairly across skin tone strata. We benchmark ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three pre-declared conditions: cross-entropy with standard augmentation, focal loss with class-balanced sampling, and cross-entropy with strong augmentation. Each condition uses three seeds, a lesion-aware split, and a validation-selected threshold at 95% specificity. Focal loss with class-balanced sampling improves ViT-S/16 by 8.2 sensitivity points but reduces ResNet-50 by 11.7 and EfficientNet-B0 by 7.9, which reverses the architecture ranking. The AUC ranking is stable. A lightness-stratified analysis shows that the darkest stratum has the lowest AUC and the highest false positive rate in all nine configurations, but only 43 test images and six melanomas fall in that stratum, so the sensitivity estimate is underpowered. Across all three architectures, Grad-CAM attribution mass inside the lesion is lower in the dark stratum than in the light stratum (ρ from -0.17 to -0.18, $p < 10^{-7}$), and pointing-game accuracy is lowest in the dark stratum. Attribution mass is a weak predictor of individual errors, so the XAI signal is group-level rather than per-image. External validation on 1,280 HIBA dermoscopic images reproduced the CNN-transformer gap but showed a 16 to 18 point sensitivity drop, and HIBA contains no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images. The results show that imbalance correction is not neutral across architectures and that dark-skin dermoscopic data remain the main bottleneck for a reliable fairness claim.

1 Introduction

Melanoma is the most lethal form of skin cancer, and survival depends on early detection. Deep learning models have reached dermatologist-level accuracy on public dermoscopic benchmarks such as HAM10000 and the ISIC challenges [13, 2]. Three problems remain.

Public datasets are imbalanced. Benign lesions outnumber melanomas by roughly ten to one, and a classifier that predicts the majority class can look accurate while missing cancers. Focal loss and class-balanced sampling are common remedies [6], but their effect is often reported for one architecture and one dataset.

The datasets are also skewed toward lighter skin. Fitzpatrick skin types V and VI are rare in public dermoscopic data, and models trained on lighter skin can lose sensitivity on darker skin [5, 3, 10]. The data needed to measure the gap are themselves scarce.

Deep models are opaque as well. Gradient-based attribution methods such as Grad-CAM and Integrated Gradients can show which pixels drive a prediction [11, 12], but most dermatology papers use them as illustrations. They are rarely turned into quantitative measurements that can be tested across skin tone groups.

Most studies examine one of these problems at a time. Architecture comparisons often apply a single imbalance correction without checking whether it is neutral across architectures, and fairness studies report group-level performance without connecting it to attribution behaviour. This study connects the three in one benchmark.

We study HAM10000 [13] with a binary melanoma endpoint and three architectures: ResNet-50, EfficientNet-B0, and ViT-S/16. We use a lesion-aware split, three seeds per condition, and a validation-selected threshold at 95% specificity. The test set is touched once per run.

Our contributions are:

1. A controlled benchmark of three architectures under three pre-declared conditions: CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

2. Evidence that focal loss with class-balanced sampling improves ViT-S/16 but reduces the sensitivity of both CNNs, which reverses the architecture ranking on sensitivity at 95% specificity. The AUC ranking is stable.

3. A lightness-stratified fairness analysis that quantifies the representation gap and a consistent dark-stratum AUC and false positive rate signal, while showing that the dark-stratum sensitivity estimate is underpowered.

4. A quantitative XAI protocol for attribution localization, used to test whether attribution mass inside the lesion is associated with lightness group and error status. All three architectures place less attribution inside the lesion on darker images, but attribution mass is a weak predictor of individual errors.

5. An external validation on HIBA that reproduces the CNN-transformer gap, shows a 16 to 18 point sensitivity drop, and confirms that HIBA contains no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images.

The training and fairness analysis use one public dataset, HAM10000. The lightness strata are image-based measurements, not clinical Fitzpatrick labels, and the results do not establish clinical utility. External validation on HIBA is reported in Section 4.6.

2 Related work

2.1 Melanoma classification and public datasets

Public dermoscopic datasets have driven most of the recent progress. HAM10000 contains 10,015 images across seven diagnostic classes [13], and the ISIC challenges have expanded the scale and the number of sources [2]. Models based on ResNet, EfficientNet, ConvNeXt, ViT, and Swin have all been reported at or above dermatologist-level accuracy on these benchmarks. The comparisons usually differ in preprocessing, split protocol, and threshold choice, which makes cross-paper rankings difficult to interpret.

2.2 Class imbalance

Focal loss down-weights easy examples and is the most common loss-level remedy for class imbalance [6]. Class-balanced sampling and class-weighted cross-entropy are common data-level and loss-level alternatives. Strong photometric and geometric augmentation is often added at the same time. The three interventions are usually evaluated together, so their separate effects and their interactions with architecture are rarely reported.

2.3 Fairness in dermatology

Fitzpatrick17k provided the first large clinical image set with Fitzpatrick labels and showed performance gaps across skin types [5]. The Diverse Dermatology Images dataset added biopsy-proven clinical images with balanced skin tone representation [3]. SCIN added crowdsourced clinical images with self-reported and estimated Fitzpatrick labels [14]. Pipsqueak collected dermoscopic melanocytic lesions in darker skin and reported a large sensitivity drop on Fitzpatrick V-VI [10]. A 2025 systematic review identifies fairness, generalizability, and replicability as the main unresolved problems in melanoma AI [8].

The dermoscopic datasets with Fitzpatrick labels remain small. HIBA contains 1,616 images from Argentina with Fitzpatrick labels [9], and PROVe-AI contains 603 biopsy-confirmed lesions [7]. Neither includes Fitzpatrick V-VI. This limits what can be claimed from dermoscopic data alone.

2.4 Explainable AI in dermatology

Grad-CAM and Integrated Gradients are the most used attribution methods in dermatology [11, 12]. Clinician-facing studies show that explanations can improve trust and diagnostic accuracy [1], but the explanations are usually evaluated qualitatively. Quantitative localization and faithfulness metrics are more common in the general XAI literature than in dermatology.

2.5 Quantitative XAI metrics

Attribution localization can be measured against lesion segmentation masks: the fraction of attribution mass inside the lesion, the pointing game, and the attribution IoU. Faithfulness can be measured with deletion and insertion curves, infidelity, and sensitivity-n. These metrics make it possible to test whether an explanation is clinically meaningful rather than visually plausible.

2.6 Gap

Three questions remain open: whether imbalance correction changes the ranking of architecture families, whether the correction affects skin tone groups differently, and whether attribution metrics can detect a group-level problem before it appears in aggregate sensitivity. We address the first two and define the protocol for the third.

3 Methods

3.1 Data and splits

We use HAM10000, which contains 10,015 dermoscopic images from 7,470 lesions across seven diagnostic classes [13]. The task is binary: melanoma versus all other diagnoses. The dataset contains 1,113 melanoma images, 11.1% of the total.

We use a lesion-aware split so that images of the same lesion never appear in more than one split. The split is 70% training, 15% validation, and 15% test. The test set contains 1,527 images and 186 melanomas. The split is generated once and reused by every run.

3.2 Preprocessing and augmentation

Images are resized to 160 pixels for the Stage 1 screen and 224 pixels for the Stage 2 comparison, and normalized with ImageNet statistics. The standard augmentation pipeline applies horizontal and vertical flips, random 90-degree rotations, and mild color jitter. The strong pipeline increases the color jitter and adds affine transformations with scale, translation, and rotation. The strong regime is designed to simulate domain shift.

3.3 Architectures

We compare ResNet-50, EfficientNet-B0, and ViT-S/16. All models use ImageNet pretrained weights and a two-class output. The ViT uses dynamic image sizing so that it accepts the 224-pixel input. The three models cover a standard CNN, an efficient CNN, and a transformer.

3.4 Training

All models use AdamW with a learning rate of 3×10^{-4}, weight decay 1×10^{-4}, and a cosine schedule. The batch size is 64, the maximum number of epochs is 15, and early stopping uses a patience of 5 on validation balanced accuracy. Gradients are clipped to a norm of 1.0. Training uses mixed precision when a GPU is available.

3.5 Conditions

Stage 1 is a screen on EfficientNet-B0 at 160 pixels. It contains eight configurations: CE with a random sampler, CE with class-balanced sampling, weighted CE with a random sampler, focal loss with $\gamma = 2$ and a random sampler, focal loss with $\gamma = 2$ and class-balanced sampling, focal loss with $\gamma = 3$ and class-balanced sampling, and two strong-augmentation variants. Each configuration uses three seeds, for 24 runs.

Stage 2 compares three pre-declared conditions across the three architectures at 224 pixels:

1. CE baseline: cross-entropy loss, random sampler, standard augmentation.
2. Planned imbalance correction: focal loss with $\gamma = 2$, class-balanced sampling, standard augmentation.
3. Stage 1 winner: cross-entropy loss, random sampler, strong augmentation.

Each condition uses three seeds, for 27 runs. The Stage 2 conditions are fixed before the test results are seen.

3.6 Metrics and threshold

We report sensitivity, specificity, balanced accuracy, AUC, and sensitivity at 95% specificity. The operating threshold is selected on the validation split at 95% specificity, frozen, and applied once to the test split. Each run writes per-sample test predictions so that the same scores can be reused for fairness and XAI analysis.

3.7 Statistical analysis

We report the mean and standard deviation across three seeds. We add 1,000-sample bootstrap confidence intervals on test for sensitivity, specificity, and AUC. The bootstrap resamples images within a run and skips degenerate samples. The intervals describe test-set uncertainty. The seed standard deviation describes run-to-run variation.

3.8 Fairness analysis

HAM10000 has no Fitzpatrick labels. We therefore compute an image-based lightness proxy from non-lesional skin. The image is converted to CIELAB, the lesion mask is dilated by 11 pixels and excluded, and very dark pixels ($L^* < 10$) and very bright pixels ($L^* > 95$) are removed. The median L^* of the remaining skin pixels is the proxy.

We group images into light ($L^* > 70$), medium ($55 < L^* \leq 70$), and dark ($L^* \leq 55$). These thresholds are specific to the HAM10000 lightness distribution. They are image-based strata, not clinical Fitzpatrick labels.

We also computed the Individual Typology Angle (ITA), $\text{ITA} = \arctan((L^* - 50)/b^*) \times 180/\pi$. The ITA was unstable on 37% of images because the b^* channel was near zero or negative after dermatoscope color casts. The L^* proxy is therefore the primary analysis, and the ITA is treated as a sensitivity check.

For each Stage 2 run, we compute per-group sensitivity, specificity, false positive rate, false negative rate, AUC, and bootstrap intervals. We report the maximum group gap and the light-to-dark sensitivity ratio.

External validation uses HIBA [9], a dermoscopic dataset from Argentina with Fitzpatrick labels. We apply the frozen HAM10000 validation threshold and do not refit on HIBA. PROVe-AI [7] is a possible extension but was not evaluated here. A clinical dark-skin dataset, such as DDI [3] or PASSION [4], is left for future work, with the imaging modality difference stated as a limitation.

3.9 XAI analysis

For the CE-baseline run of each architecture, we compute Grad-CAM heatmaps on a tone-stratified test sample: all 43 dark images plus 150 randomly drawn light and 150 medium images per seed. We measure:

- Attribution mass inside the lesion: $M = \sum_{i \in \text{lesion}} h_i / \sum_i h_i$, where h_i is the normalized heatmap value at pixel i.
- Pointing game accuracy: the fraction of images where the maximum heatmap value falls inside the lesion mask.
- Attribution IoU: the intersection over union between the thresholded heatmap and the lesion mask.

We then compute the Spearman correlation between attribution mass and error status, and between attribution mass and lightness group. The correlations pool the three seeds, so the p-values treat repeated images as independent and are descriptive rather than confirmatory. The XAI analysis uses the same test split and the same per-sample predictions as the classification and fairness analyses.

We also implemented deletion and insertion curves, but the models are highly confident: the target-class probability stays near 1.0 for both the original and the perturbed images, so the curves

Table 1: Stage 1 screen on EfficientNet-B0 at 160 pixels. Validation sensitivity is used for model selection. Test sensitivity is reported at the validation-selected 95% specificity threshold. Values are mean and standard deviation across three seeds.

Configuration	Val sens	Test sens (95% CI)	Specificity	Balanced acc	AUC
stage1_ce_random_strong	0.556 ± 0.027	0.513 (0.438 to 0.582)	0.943	0.728	0.901
stage1_ce_random	0.536 ± 0.016	0.530 (0.456 to 0.600)	0.948	0.739	0.897
stage1_ce_cb	0.521 ± 0.086	0.484 (0.411 to 0.554)	0.938	0.711	0.890
stage1_focal2_random	0.509 ± 0.041	0.487 (0.412 to 0.560)	0.946	0.717	0.894
stage1_focal2_cb	0.502 ± 0.039	0.480 (0.408 to 0.551)	0.943	0.712	0.897
stage1_wce_random	0.502 ± 0.043	0.493 (0.421 to 0.564)	0.937	0.715	0.896
stage1_focal2_cb_strong	0.483 ± 0.013	0.491 (0.421 to 0.558)	0.943	0.717	0.896
stage1_focal3_cb	0.449 ± 0.051	0.478 (0.405 to 0.553)	0.946	0.712	0.899

do not separate heatmaps on this benchmark. We therefore report localization metrics and treat faithfulness curves as future work.

3.10 Reproducibility

Every run writes its configuration, git commit, seed, device, package versions, metrics, and per-sample predictions. The split is cached and reused. The code, configs, and table generation scripts are available in the project repository.

4 Results

4.1 Stage 1: loss, sampling, and augmentation screen

Table 1 reports the Stage 1 screen on EfficientNet-B0 at 160 pixels. The validation winner is CE with strong augmentation, with a validation sensitivity of 0.556 ± 0.027 at 95% specificity. The CE baseline reaches 0.536 ± 0.016. The planned focal loss with class-balanced sampling reaches 0.502 ± 0.039, and focal loss with $\gamma = 3$ and class-balanced sampling reaches 0.449 ± 0.051.

The planned imbalance correction did not beat the CE baseline on validation. The difference between the CE baseline and the strong-augmentation variant is small relative to the seed standard deviation. The test columns in Table 1 are descriptive; the model selection used validation only.

4.2 Stage 2: architecture comparison

Table 2 reports the Stage 2 comparison at 224 pixels. The three conditions are the CE baseline, focal loss with class-balanced sampling, and CE with strong augmentation.

At the CE baseline, ResNet-50 reaches a test sensitivity of 0.570 (95% CI 0.497 to 0.639), EfficientNet-B0 reaches 0.556 (0.480 to 0.624), and ViT-S/16 reaches 0.400 (0.329 to 0.471). The two CNNs lead the transformer by a wide margin.

Under focal loss with class-balanced sampling, ResNet-50 falls to 0.453 (0.379 to 0.523), EfficientNet-B0 falls to 0.477 (0.405 to 0.548), and ViT-S/16 rises to 0.482 (0.408 to 0.556). The correction helps the transformer by 8.2 points and hurts the CNNs by 11.7 and 7.9 points.

The architecture ranking therefore reverses. At the CE baseline the order is ResNet-50, EfficientNet-B0, ViT-S/16. Under focal loss with class-balanced sampling the order is ViT-S/16, EfficientNet-B0, ResNet-50.

Table 2: Stage 2 comparison at 224 pixels. Test sensitivity is reported at the validation-selected 95% specificity threshold. Values are mean and standard deviation across three seeds.

Configuration	Val sens	Test sens (95% CI)	Specificity	Balanced acc	AUC
stage2_efficientnet_b0_ce_random_strong	0.521 ± 0.021	0.563 (0.489 to 0.636)	0.952	0.757	0.912
stage2_efficientnet_b0_ce_random	0.553 ± 0.035	0.556 (0.480 to 0.624)	0.953	0.754	0.914
stage2_efficientnet_b0_focal2_cb	0.481 ± 0.042	0.477 (0.405 to 0.548)	0.950	0.713	0.891
stage2_resnet50_ce_random	0.526 ± 0.044	0.570 (0.497 to 0.639)	0.951	0.761	0.897
stage2_resnet50_ce_random_strong	0.549 ± 0.015	0.568 (0.498 to 0.637)	0.942	0.755	0.899
stage2_resnet50_focal2_cb	0.489 ± 0.027	0.453 (0.379 to 0.523)	0.947	0.700	0.884
stage2_vit_s_focal2_cb	0.474 ± 0.034	0.482 (0.408 to 0.556)	0.940	0.711	0.872
stage2_vit_s_ce_random_strong	0.466 ± 0.030	0.435 (0.366 to 0.503)	0.941	0.688	0.866
stage2_vit_s_ce_random	0.423 ± 0.042	0.400 (0.329 to 0.471)	0.941	0.670	0.854

Strong augmentation is close to neutral. ResNet-50 reaches 0.568 (0.498 to 0.637), EfficientNet-B0 reaches 0.563 (0.489 to 0.636), and ViT-S/16 reaches 0.435 (0.366 to 0.503). The intervals overlap with the CE baseline for the two CNNs.

4.3 AUC is stable while sensitivity ranking flips

The AUC ranking does not follow the sensitivity ranking. At the CE baseline, the AUC is 0.914 for EfficientNet-B0, 0.897 for ResNet-50, and 0.854 for ViT-S/16. Under focal loss with class-balanced sampling, the AUC is 0.891 for EfficientNet-B0, 0.884 for ResNet-50, and 0.872 for ViT-S/16. EfficientNet-B0 remains first and ViT-S/16 remains last in both conditions.

The stability of the AUC ranking suggests that the sensitivity reversal is specific to the thresholded operating point. Calibration and threshold placement are plausible mechanisms. Both metrics matter: a clinical screening claim depends on the thresholded sensitivity, and the AUC describes the ranking quality of the scores.

4.4 Fairness across lightness strata

The test set contains 644 light images with 81 melanomas, 840 medium images with 99 melanomas, and 43 dark images with 6 melanomas. The dark stratum is 2.8% of the test set and 3.2% of the melanoma cases.

Table 3 reports the per-group sensitivity and the dark-stratum AUC and false positive rate for the nine Stage 2 configurations. Figure 1 shows the AUC and false positive rate pattern.

The dark stratum has the lowest AUC in all nine configurations. The dark AUC ranges from 0.71 to 0.87, against 0.83 to 0.90 for the light stratum and 0.88 to 0.93 for the medium stratum.

It also has the highest false positive rate. The dark false positive rate ranges from 0.08 to 0.19, against 0.04 to 0.07 for the light and medium strata.

The dark-stratum sensitivity is too uncertain to support a strong claim. With six melanoma cases, the estimates range from 0.22 to 0.72 and the bootstrap intervals span 0 to 1.

The medium stratum has the highest sensitivity in most configurations. The light stratum is often lower, which is consistent with a contrast effect: pale lesions on pale skin are harder to separate. Lesion contrast, not only skin tone, appears to drive difficulty.

4.5 Quantitative XAI

Table 4 reports attribution mass inside the lesion, pointing-game accuracy, attribution IoU, and error rate for the CE-baseline models. Table 5 reports the Spearman correlations. The sample

Table 3: Lightness-stratified test sensitivity at 95% specificity. Values are mean and standard deviation across three seeds. The dark stratum contains 43 test images and six melanoma cases. These are image-based lightness groups, not clinical Fitzpatrick labels.

Configuration	Light sens	Medium sens	Dark sens	Dark AUC	Dark FPR	Dark n / pos.
EfficientNet-B0 / CE	0.477 ± 0.031	0.609 ± 0.050	0.722 ± 0.096	0.871	0.117	43 / 6
EfficientNet-B0 / CE+strong	0.519 ± 0.054	0.599 ± 0.086	0.556 ± 0.096	0.833	0.090	43 / 6
EfficientNet-B0 / Focal+CB	0.420 ± 0.064	0.522 ± 0.076	0.500 ± 0.167	0.806	0.081	43 / 6
ResNet-50 / CE	0.502 ± 0.036	0.633 ± 0.050	0.444 ± 0.096	0.803	0.117	43 / 6
ResNet-50 / CE+strong	0.527 ± 0.026	0.603 ± 0.038	0.556 ± 0.096	0.854	0.135	43 / 6
ResNet-50 / Focal+CB	0.399 ± 0.093	0.481 ± 0.066	0.722 ± 0.192	0.848	0.162	43 / 6
ViT-S/16 / CE	0.342 ± 0.091	0.451 ± 0.065	0.333 ± 0.167	0.740	0.144	43 / 6
ViT-S/16 / CE+strong	0.407 ± 0.054	0.471 ± 0.132	0.222 ± 0.096	0.763	0.126	43 / 6
ViT-S/16 / Focal+CB	0.362 ± 0.062	0.586 ± 0.010	0.389 ± 0.096	0.712	0.189	43 / 6

contains 129 dark images and 18 melanomas across the three seeds, and 450 light and 450 medium images.

Attribution mass inside the lesion is highest in the light stratum for all three architectures. EfficientNet-B0 falls from 0.456 to 0.374 to 0.342 across light, medium, and dark. ViT-S/16 falls from 0.341 to 0.251 and then rises slightly to 0.265. ResNet-50 falls from 0.593 to 0.501 and then stays near 0.504. The mass-tone correlation is negative and significant for every architecture: -0.170 for EfficientNet-B0, -0.167 for ResNet-50, and -0.184 for ViT-S/16, all with $p < 10^{-7}$.

Pointing-game accuracy is lowest in the dark stratum in every case: 0.449 for EfficientNet-B0, 0.674 for ResNet-50, and 0.248 for ViT-S/16. Attribution IoU follows the same direction for EfficientNet-B0 and ViT-S/16, and is only slightly lower in the dark stratum than the medium stratum for ResNet-50 (0.330 vs 0.362).

The error rate is higher in the dark stratum than in the light stratum for every architecture: 0.140 vs 0.076 for EfficientNet-B0, 0.178 vs 0.076 for ResNet-50, and 0.217 vs 0.109 for ViT-S/16. The direction matches the fairness results, though the dark sample is small.

Attribution mass does not predict individual errors well. The mass-error correlation is -0.023 for EfficientNet-B0 ($p = 0.46$), 0.047 for ResNet-50 ($p = 0.13$), and 0.089 for ViT-S/16 ($p = 0.004$). Only the ViT correlation is significant, and its effect size is small. The XAI signal is therefore group-level: all three models place less attribution inside the lesion on darker images, and darker images have higher error rates. It does not identify which individual images will fail.

Figure 2 shows the group pattern. ResNet-50 localizes best in absolute terms and has the highest dark-stratum attribution mass, but its dark error rate is higher than EfficientNet-B0's. Localization quality and error rate move together across tone groups, but they do not rank architectures the same way.

4.6 External tone-labeled validation

Table 6 reports the HIBA dermoscopic subset: 1,280 images and 196 melanomas. We applied the HAM10000 validation threshold at 95% specificity without refitting. HIBA contains Fitzpatrick I-IV only, and 108 images have no Fitzpatrick label.

Sensitivity drops on the external set. ResNet-50 reaches 0.391 at 0.944 specificity, EfficientNet-B0 reaches 0.396 at 0.922, and ViT-S/16 reaches 0.221 at 0.967. The AUCs are 0.807, 0.782, and 0.791. On the HAM10000 test set the same models reached 0.57, 0.56, and 0.40 sensitivity at 95% specificity, with AUCs of 0.90, 0.91, and 0.85. Both CNNs outperform the transformer on external sensitivity, which matches the HAM10000 baseline, but the two CNNs are within half a

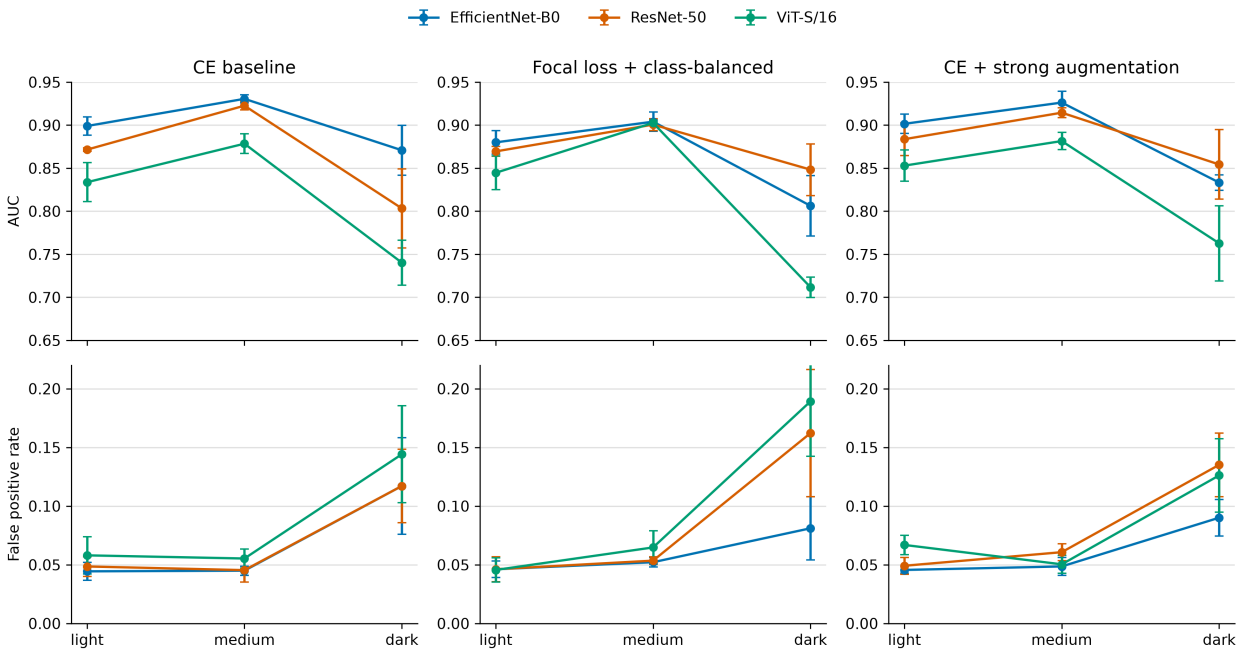


Figure 1: AUC and false positive rate by lightness group for the nine Stage 2 configurations. Columns are training conditions and lines are architectures. Points are means across three seeds; error bars are seed standard deviations. The dark stratum contains 43 test images and six melanoma cases.

point of each other. The absolute drop is consistent with domain shift between the Argentine HIBA collection and HAM10000.

The per-type estimates do not show a monotonic skin-tone gradient. For Fitzpatrick I, II, and III, CNN sensitivity ranges from 0.35 to 0.43, and the ViT is lower at 0.20 to 0.33. Fitzpatrick IV contains 25 images and no melanomas, so sensitivity is undefined and only specificity can be estimated. The group sizes are unbalanced: 804 of the 1,280 images are type II. HIBA cannot test the dark-skin claim, because it has almost no type IV melanoma cases and no type V-VI images at all.

The CNN-transformer gap transfers, so the transformer deficit is not specific to HAM10000. The absolute performance does not transfer: the frozen HAM10000 threshold yields much lower sensitivity on HIBA, which means the operating point needs recalibration for a new source. The tone-specific claim remains limited to the image-based lightness strata in HAM10000 and to the consistent dark-stratum AUC and false positive rate signal.

4.7 Figures

Figure 3 shows the training curves for the median seed of the best Stage 2 configuration by mean test sensitivity, ResNet-50 with the CE baseline. Figure 4 shows the corresponding test ROC curve.

Table 4: Quantitative XAI at the CE baseline on the HAM10000 test split. Each architecture uses three seeds and a tone-stratified sample of 343 test images per seed. Attribution mass is the fraction of the Grad-CAM heatmap inside the lesion mask. Values are means across seeds.

Architecture	Tone group	n	n positive	Attribution mass	Pointing game	Attribution IoU	Error rate
ResNet-50	light	450	36	0.593	0.882	0.404	0.076
	medium	450	54	0.501	0.809	0.362	0.078
	dark	129	18	0.504	0.674	0.330	0.178
EfficientNet-B0	light	450	36	0.456	0.718	0.333	0.076
	medium	450	54	0.374	0.613	0.284	0.091
	dark	129	18	0.342	0.449	0.208	0.140
ViT-S/16	light	450	36	0.341	0.604	0.277	0.109
	medium	450	54	0.251	0.367	0.173	0.116
	dark	129	18	0.265	0.248	0.154	0.217

Table 5: Spearman correlations between attribution mass inside the lesion, lightness group, and error status at the CE baseline. Tone is coded light = 0, medium = 1, dark = 2. A negative mass-tone correlation means less attribution inside the lesion on darker images.

Architecture	Images	ρ (mass, tone)	p	ρ (mass, error)	p
EfficientNet-B0	1027	-0.170	4.6×10^{-8}	-0.023	0.458
ResNet-50	1029	-0.167	7.3×10^{-8}	0.047	0.131
ViT-S/16	1029	-0.184	3.0×10^{-9}	0.089	0.004

5 Discussion

5.1 Imbalance correction is architecture-dependent

Focal loss with class-balanced sampling is not a neutral intervention. It helps the transformer and hurts both CNNs. A benchmark that applied the correction to one architecture and generalized the result would have reached the wrong conclusion about the others.

The effect has a plausible mechanism. Focal loss changes the gradient distribution, and class-balanced sampling changes the effective class prior. A transformer with weaker inductive bias may benefit from the harder examples and the reweighted prior, while a CNN with strong local features may overfit the minority samples or lose calibration. The experiment cannot separate those mechanisms. We report the effect as an empirical interaction, not a causal explanation.

5.2 The ranking flip and the stable AUC

The sensitivity ranking reverses, but the AUC ranking does not. EfficientNet-B0 remains first and ViT-S/16 remains last in both the baseline and corrected conditions. The reversal is therefore specific to the thresholded operating point.

This matters for how model comparisons are reported. A single threshold at 0.5, or a single sensitivity number, can make an architecture look better or worse depending on the correction. Reporting sensitivity at a fixed specificity, AUC, and calibration together gives a more complete picture. We leave a full calibration analysis to future work.

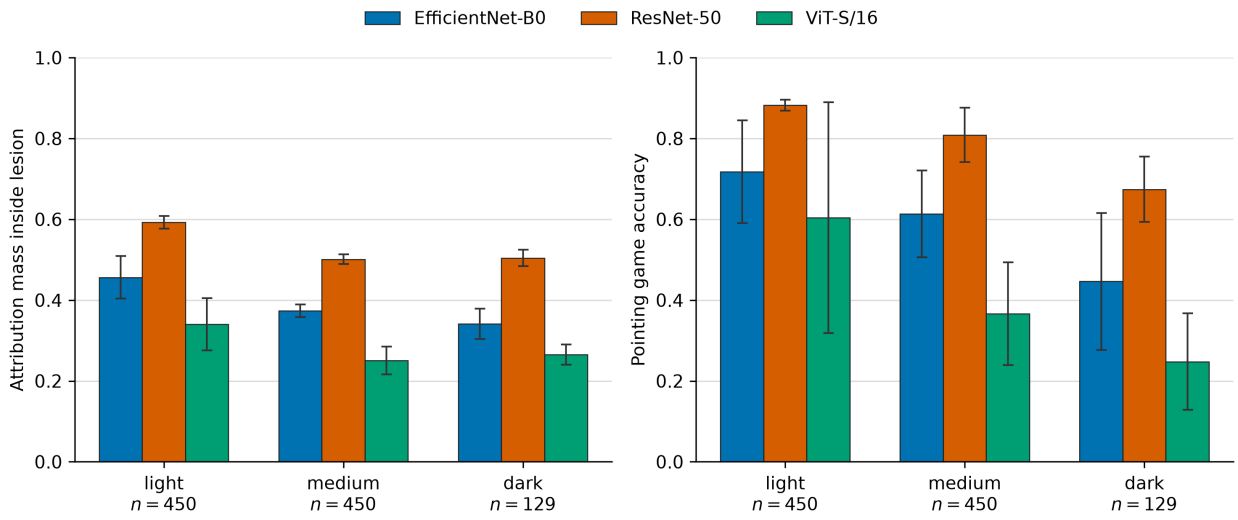


Figure 2: Attribution mass inside the lesion and pointing-game accuracy by lightness group for the three CE-baseline architectures. Bars are means across three seeds; error bars are seed standard deviations. The sample contains all 43 dark test images per seed and 150 randomly drawn light and medium images per seed.

5.3 The fairness signal

The dark stratum has the lowest AUC and the highest false positive rate in all nine configurations. The direction is consistent even though the sample is small. The most likely interpretation is that the model's ranking quality degrades on darker images, which produces more false positives at the fixed specificity threshold.

The dark-stratum sensitivity is underpowered. Six melanoma cases cannot support a precise estimate, and the bootstrap intervals span the full range. We therefore treat the AUC and false positive rate as the more stable signals and the sensitivity estimate as inconclusive.

The light stratum is often harder than the medium stratum. This is consistent with a contrast effect: a pale lesion on pale skin has a lower boundary contrast, and the model may attend to the surrounding skin. Skin tone is not the only source of difficulty. Lesion contrast, hair, rulers, and illumination can all change the model's attention.

5.4 The data bottleneck

The external dermoscopic datasets with Fitzpatrick labels, HIBA and PROVe-AI, contain Fitzpatrick I-IV only. The HIBA evaluation makes the constraint concrete. In 1,280 dermoscopic images, Fitzpatrick IV accounts for 25 images and not a single melanoma. Sensitivity is estimable only for types I-III, and those estimates show no monotonic skin-tone gradient. The 16 to 18 point sensitivity drop between HAM10000 and HIBA also shows that the available external set is a different imaging domain, not a clean replication.

The clinical datasets with Fitzpatrick V-VI, such as DDI, SCIN, and PASSION, differ from HAM10000 in imaging modality. This is the main obstacle to a strong dark-skin claim. This study can quantify the representation gap, report the consistent dark-stratum AUC and false-positive signal, and show the limitation of the available dermoscopic data. A precise dark-skin sensitivity estimate requires new dermoscopic data collection or a carefully controlled cross-modality study.

Table 6: External validation on the HIBA dermoscopic subset (1,280 images, 196 melanomas). The threshold is the HAM10000 validation threshold at 95% specificity, frozen before the external evaluation. Values are means across three seeds. Fitzpatrick IV contains 25 images and no melanomas, so sensitivity is undefined. The 108 images without a Fitzpatrick label are included in the All row only.

Architecture	Fitzpatrick	n	Melanomas	Sensitivity	Specificity	AUC
ResNet-50	I	97	25	0.360	0.954	0.679
	II	804	144	0.403	0.922	0.780
	III	246	17	0.412	0.978	0.946
	IV	25	0	–	1.000	–
	All	1280	196	0.391	0.944	0.807
EfficientNet-B0	I	97	25	0.427	0.940	0.741
	II	804	144	0.389	0.890	0.745
	III	246	17	0.353	0.974	0.860
	IV	25	0	–	0.987	–
	All	1280	196	0.396	0.922	0.782
ViT-S/16	I	97	25	0.200	0.995	0.773
	II	804	144	0.227	0.950	0.777
	III	246	17	0.333	0.994	0.849
	IV	25	0	–	1.000	–
	All	1280	196	0.221	0.967	0.791

5.5 What XAI adds

The XAI results show a group-level pattern that the aggregate metrics do not capture. All three architectures place less attribution inside the lesion on darker images, and the dark stratum has the highest error rate in every architecture. The direction is consistent across a standard CNN, an efficient CNN, and a transformer, so it is not a quirk of one inductive bias.

The same analysis sets a limit on what attribution can claim. Attribution mass does not predict individual errors well: the correlation is near zero for both CNNs and small for the ViT. A deployment system should not use this heatmap score as a per-image triage signal without further work. It can use the group-level pattern as a monitoring signal: when a cohort shows lower attribution mass inside the lesion, the model is likely to make more errors on that cohort.

ResNet-50 is the counterexample to a simple assumption that more attribution inside the lesion means fewer errors. It has the highest attribution mass and pointing-game accuracy in the dark stratum, but its dark error rate is higher than EfficientNet-B0's. Heatmaps are a partial view of model behaviour, not a substitute for group-level performance measurement.

5.6 Implications for model selection

The choice of architecture depends on the correction. At the CE baseline, ResNet-50 has the highest sensitivity at 95% specificity. Under focal loss with class-balanced sampling, ViT-S/16 is the best. A deployment decision that ignores the training condition can pick the wrong model.

For the fairness analysis, this means the evaluation should include the architecture-condition

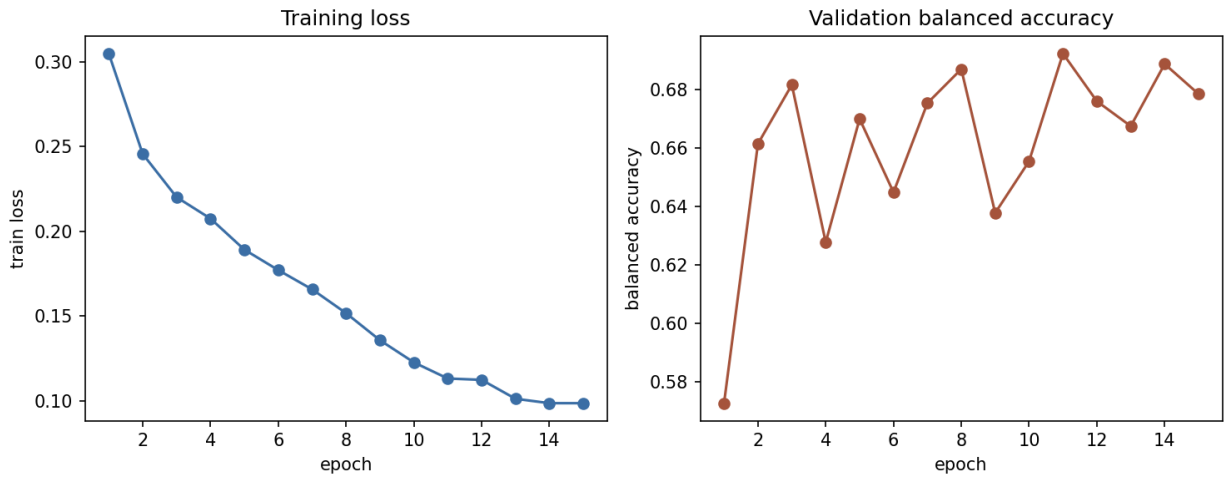


Figure 3: Training loss and validation balanced accuracy for the median seed of ResNet-50 with the CE baseline. The model trains for 15 epochs and restores the best validation checkpoint.

combinations, not a single global best model. The best model on aggregate sensitivity may not be the best model in the dark stratum, and the ranking may change again after the external validation.

6 Limitations

- We use HAM10000 only. Cross-dataset generalization to PH2, BCN20000, or ISIC 2020 is not yet measured.
- The endpoint is binary melanoma versus all other diagnoses. Seven-class classification is a secondary experiment.
- Each condition uses three seeds. The seed standard deviation is reported, but more seeds would tighten the estimates.
- Stage 1 uses 160-pixel images and Stage 2 uses 224-pixel images. The two stages are not directly comparable.
- The fairness analysis uses an image-based lightness proxy, not clinical Fitzpatrick labels. The proxy is sensitive to illumination, color calibration, and lesion contrast.
- The dark stratum contains 43 test images and six melanomas. The sensitivity estimate is underpowered, and the dark-stratum results are a signal rather than a precise effect size.
- HIBA and PROVe-AI contain Fitzpatrick I-IV only. The clinical datasets with Fitzpatrick V-VI differ in imaging modality, so the dark-skin comparison is confounded.
- Deletion and insertion curves saturated at a target probability near 1.0, so they could not separate heatmaps. Faithfulness is therefore not measured, and the XAI results cover localization only.
- No clinician reader study was conducted, so the clinical relevance of the attribution maps is not validated with dermatologists.

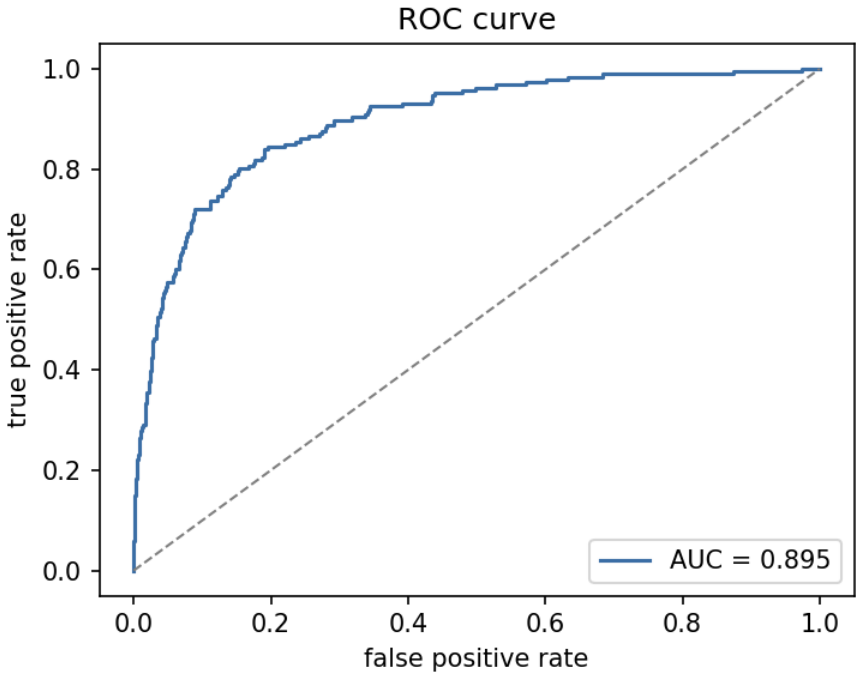


Figure 4: Test ROC curve for the median seed of ResNet-50 with the CE baseline. The operating threshold is selected on validation at 95% specificity.

- No deployed system, prospective validation, or regulatory analysis is included.

7 Ethics and reproducibility

All data are public, de-identified research datasets. HAM10000 and the ISIC archive are released for research use, and the external datasets are used under their respective licenses. The study involves no patient contact and no new data collection.

The system is a research prototype, not a diagnostic device. The results do not establish clinical utility.

The Fitzpatrick scale has known limitations: it is a coarse phototype scale, it does not capture the full range of skin tones, and it conflates pigmentation with UV response. We use it because the datasets and the fairness literature use it. The lightness strata are image-based measurements, not clinical labels.

The representation gap is a finding with an ethical consequence: the data needed to measure dark-skin performance are scarce. We report the small number of dark-skin dermoscopic images. Synthetic dark-skin images are not used to claim fairness improvements.

The code, configs, splits, per-sample predictions, and table generation scripts are released with the paper. The repository records the git commit, seed, device, and package versions for every run.

8 Conclusion

We benchmarked ResNet-50, EfficientNet-B0, and ViT-S/16 on HAM10000 under three predeclared conditions. Focal loss with class-balanced sampling improved the transformer but reduced the sensitivity of both CNNs, which reversed the architecture ranking on sensitivity at 95% specificity. The AUC ranking stayed stable, which points to calibration and threshold effects rather than a change in ranking quality.

The fairness analysis showed that the darkest lightness stratum had the lowest AUC and the highest false positive rate in all nine configurations. The stratum contains only 43 test images and six melanomas, so the sensitivity estimate is underpowered. External validation on 1,280 HIBA dermoscopic images reproduced the CNN-transformer gap but showed a 16 to 18 point sensitivity drop, and HIBA contains no Fitzpatrick IV melanomas and no Fitzpatrick V-VI images.

The XAI analysis showed that all three architectures place less attribution inside the lesion on darker images. Attribution mass was a weak predictor of individual errors, so the signal is useful for group-level monitoring rather than per-image triage. The next steps are a clinical dark-skin evaluation with DDI or PASSION and a temperature-scaled faithfulness analysis.

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